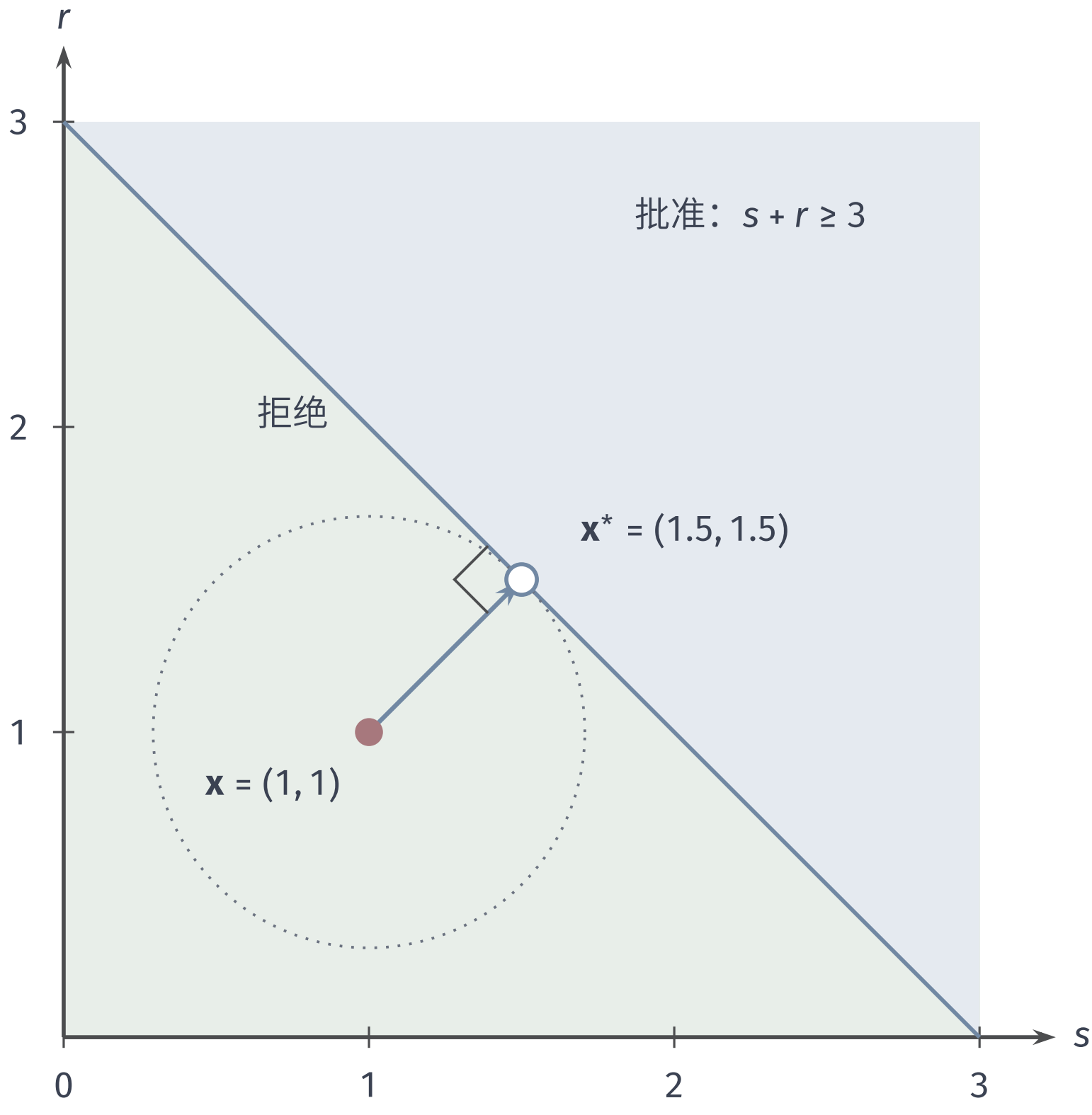

$$\|\mathbf{x}^* - \mathbf{x}\|_2 = 1/\sqrt{2}$$